In many sectors in Mathematics, the 'Invariant' objects are quite important.

[Def. Let K/F be a finite separable extension, and let L/F be Galois closure. Define

$$N_{K/F}: K \to F;\ \alpha \mapsto \prod_{\sigma \in Emb_F(K,L)} \sigma(\alpha)$$

$$Tr_{K/F}: K \to F;\ \alpha \mapsto \sum_{\sigma \in Emb_F(K,L)} \sigma(\alpha).$$

In particular, if K/F is Galois, then $L=K$, so that $Emb_F(K/L) = Gal(K/F)$.

[Prop.

1). $N_{K/F}$ and $Tr_{K/F}$ are well-defined. (That is, for any $\alpha \in K$, $N_{K/F}(\alpha), Tr_{K/F}(\alpha) \in F$)

2). The $N_{K/F}$ preserves multiplication and the $Tr_{K/F}$ preserves addition.

Proof. Let

[Def. Let E/F be a finite extension of a field F.
Given $\alpha \in E$, a map $m_\alpha : E \to E : x \mapsto \alpha x$ is F-linear map.
Define the Norm of E over F is

$$N_{E/F} : E \to F : \alpha \mapsto \det m_\alpha$$

and the Trace of E over F is

$$Tr_{E/F} : E \to F : \alpha \mapsto tr\, m_\alpha.$$

Note: Tr is an additive Homomorphism on E over F, (also linear functional)
and Norm is a multiplicative Homomorphism on $E^\times$ to $F^\times$

[Prop. Let E/F be a finite extension, and let $\alpha \in E$.
Let $P(x) \in F[x]$ be the minimal polynomial of α over F.
Put $d = \deg P = [F(\alpha) : F]$ and $S = [E : F(\alpha)]$.
Suppose that P factors over $\overline{F}$ as

$$P = \prod_{i=1}^{d} (x - \alpha_i) \quad \text{where } \alpha_1, \cdots, \alpha_d \in \overline{F}$$

Then, the characteristic polynomial of m_α is P^S, and particularly

$$N_{E/F}(\alpha) = \prod_{i=1}^{d} \alpha_i^S \quad \text{and} \quad Tr_{E/F}(\alpha) = S \cdot \sum_{i=1}^{d} \alpha_i.$$

Proof. Let $\{\beta_1, \cdots, \beta_S\}$ be a basis for E over $F(\alpha)$. Since $\{\alpha^0, \cdots, \alpha^{d-1}\}$ is a basis for $F(\alpha)$ over F, $\mathcal{B} = \{\beta_i \cdot \alpha^j \mid 1 \le i \le S,\ 0 \le j \le d-1\}$ is a basis for E over F. Consider $\mathcal{B}$ is equipped the lexicographical ordering, then

$$[m_\alpha]_{\mathcal{B}} = \begin{bmatrix} [m_\alpha]_A & & \\ & \ddots & \\ & & [m_\alpha]_A \end{bmatrix}$$

where $m_\alpha : F(\alpha) \to F(\alpha)$ & $F(\alpha)$ is m_α invariant.
and $A := \{\alpha^0, \cdots, \alpha^{d-1}\}$.
($[m_\alpha]_A$ is a companion matrix for P!)

$\beta_1 \cdot A \quad \cdots \quad \beta_S \cdot A$

Therefore, the chariteristic polynomial of m_α is P^S. And,

$$P^S(x) = x^d - tr(m_\alpha) \cdot x^{d-1} + \cdots + (-1)^d \cdot \det(m_\alpha)$$

gives the result. □

$$\det(xI_n - A) = \sum_{\sigma \in S_n} \operatorname{sgn} \sigma \cdot$$

Hilbert 90.

Thm. Let E/F be a finite cyclic extension, and let σ be a generator of $Gal(E/F)$. Then,

$$\ker N_{E/F} = \left\{ \frac{\sigma(\beta)}{\beta} \,\middle|\, \beta \in E^\times \right\}.$$

Proof. Put $n = [E:F]$. Let $\beta \in E^\times$ be given. Then,

$$N_{E/F}\left(\frac{\sigma(\beta)}{\beta}\right) = \prod_{i=0}^{n-1} \sigma^i\left(\frac{\sigma(\beta)}{\beta}\right) = \prod_{i=0}^{n-1} \frac{\sigma^{i+1}(\beta)}{\sigma^i(\beta)} = 1.$$

So, the $\supseteq$ inclusion is shown. Conversely, let $\alpha \in \ker N_{E/F}$ be given. For each $\gamma \in E$, let

$$\begin{aligned} x_\gamma &:= \gamma + \alpha\cdot\sigma(\gamma) + \alpha\cdot\sigma(\alpha)\cdot\sigma^2(\gamma) + \cdots + \alpha\,\sigma(\alpha)\cdots\sigma^{n-2}(\alpha)\cdot\sigma^{n-1}(\gamma) \\ &= \sum_{i=0}^{n-1}\left(\prod_{j=0}^{i-1}\sigma^j(\alpha)\right)\cdot\sigma^i(\gamma) \end{aligned}$$

Since $Gal(E/F) = \langle\sigma\rangle$ is E-linearly independent, there exists $\gamma \in E^\times$ such that $x_\gamma \neq 0$. (If for all $\gamma \in E^\times$, $x_\gamma = 0$ then $\alpha = 0$, it is contradiction.) And,

$$\begin{aligned} \alpha\cdot\sigma(x_\gamma) &= \alpha\cdot\sigma\left(\sum_{i=0}^{n-1}\left(\prod_{j=0}^{i-1}\sigma^j(\alpha)\right)\cdot\sigma^i(\gamma)\right) \\ &= \alpha\cdot\sum_{i=0}^{n-1}\left(\prod_{j=0}^{i-1}\sigma^{j+1}(\alpha)\right)\cdot\sigma^{i+1}(\gamma) = x_\gamma, \text{ so that} \end{aligned}$$

$$\alpha = \frac{x_\gamma}{\sigma(x_\gamma)} = \frac{\sigma(x_\gamma^{-1})}{x_\gamma^{-1}}.$$